My Python Program Collection

1. Basic Programs & Variables

hello.py

```
num1 = int(input("Enter first number: "))
num2 = int(input("Enter second number: "))
result = num1 + num2
print(result)
```

A.py

```
value = int(input("Enter the limit: "))
total = 0

for number in range(1, value + 1):
    total = total + number

print("Total =", total)
```

mov.py

```
weight = float(input("Enter mass in kg: "))
speed = float(input("Enter velocity in m/s: "))

moment = weight * speed

print("Momentum is:", moment)
```

MOVM.py

```
birth = int(input("Enter birth year: "))
income = float(input("Enter salary in Rs: "))

current_age = 2025 - birth
usd = income / 83

print(current_age)
print(usd)
print("<<<<<<<<<<< END >>>>>>>>>>>")
```

m.py

```
mass_value = float(input("Enter mass in kg: "))
velocity_value = float(input("Enter velocity in m/s: "))

energy = mass_value * velocity_value * velocity_value
print("Calculated value is:", energy)
```

3.py

```
n1 = int(input())
n2 = int(input())
n3 = int(input())
n4 = int(input())
n5 = int(input())

print("Largest number:", max(n1, n2, n3, n4, n5))
```

loop.py

```
count = 1
while count <= 5:
    print(count)
```

loop1.py

```
number = 15
price = 8.5
message = "Python"
status = False

print(type(number))
print(type(price))
print(type(message))
print(type(status))
```

inc2.py

```
for value in range(15, 2, -3):
    print(value)
```

---

2. Scope — Local, Global & Nonlocal

local.py

```
def display_value():
    local_num = 25
    print(local_num)

display_value()
print(local_num)
```

global.py

```
count = 100

def print_value():
    print(count)

print_value()
print(count)

print("NEXT")

count = 30

def modify():
    global count
    count = 60

modify()
print(count)
```

5.py

```
def sample():
    data = 50
    print("Inside:", data)

sample()
print("Outside:", data)
```

6.py

```
value = 7

def demo():
    value = 14
```

```
    print("Inside Function:", value)

demo()
print("Outside Function:", value)

print("============================")

value = 40

def show_data():
    print("Inside:", value)

show_data()
print("Outside:", value)

print("============================")

value = 12

def update_data():
    global value
    value = 22

update_data()
print("Outside:", value)
```

7.py

```
def main_function():
    number = 300

    def inner_function():
        nonlocal number
        number = 500

    inner_function()
    print("Outer Value:", number)

main_function()
```

nonlocal.py

```
def parent():
    amount = 15

    def child():
        nonlocal amount
        amount = 35
        print("Inner:", amount)
```

```
    child()
    print("Outer:", amount)

parent()
```

---

3. Strings & Type Conversion

string.py

```
monthly_salary = []

months = int(input("Enter number of months: "))

for item in range(months):
    salary_amount = int(input(f"Enter salary for month {item + 1}: "))
    monthly_salary.append(salary_amount)

print("Total:", sum(monthly_salary))
print("Average:", sum(monthly_salary) / months)
print("Highest:", max(monthly_salary))
print("Lowest:", min(monthly_salary))
```

split.py

```
text = input("Enter sentence: ")

parts = text.split()

print("Words:")
print(parts)

print("Second Example")

values = input("Enter marks separated using comma: ")

mark_list = values.split(",")

print("Marks are:")
print(mark_list)

print("Another Example")

line = input("Enter sentence: ")
word_count = line.split()
```

```
print("Word Count:", len(word_count))
```

vowel.py

```
number = 20
print("Before Conversion:", type(number))

number = float(number)
print("After Conversion:", type(number))
```

---

4. Conditions & Loops

if.py

```
number = int(input("Enter a value: "))

for table in range(1, 11):
    print(number * table)

table = table + 1
```

leap.py

```
year_value = int(input("Enter year: "))

if (year_value % 4 == 0 and year_value % 100 != 0) or (year_value % 400 ==
0):
    print("Leap Year")
else:
    print("Not Leap Year")
```

assign.py

```
limit = int(input("Enter limit: "))
answer = 0

for num in range(1, limit + 1):
    answer = answer + num

print("Answer =", answer)

print("--------------------------------")

check_num = int(input("Enter number: "))
```

```
if check_num % 3 == 0 and check_num % 5 == 0:
    print("Divisible by 3 and 5")
else:
    print("Not divisible")
```

---

5. Functions

lambda1.py

```
check_even = lambda value: value % 2 == 0
print(check_even(10))
```

evenodd1.py

```
def calculate_sum(start_num, end_num):
    even_total = 0
    odd_total = 0

    for number in range(start_num, end_num + 1):
        if number % 2 == 0:
            even_total = even_total + number
        else:
            odd_total = odd_total + number

    return even_total, odd_total


begin = int(input("Enter starting value: "))
finish = int(input("Enter ending value: "))

output = calculate_sum(begin, finish)

print("Even Sum:", output[0])
print("Odd Sum:", output[1])
```

bonus.py

```
def bonus_calculation(pay, experience, performance):
    if pay >= 50000 and experience >= 5 and performance >= 4:
        incentive = pay * 0.20
        print("High Bonus =", incentive)

    elif pay >= 30000 and experience >= 3 and performance >= 3:
        incentive = pay * 0.10
        print("Medium Bonus =", incentive)
```

```
    elif pay >= 20000 and experience >= 1 and performance >= 2:
        incentive = pay * 0.05
        print("Low Bonus =", incentive)

    else:
        print("No Bonus")


salary = float(input("Enter salary: "))
years = int(input("Enter years: "))
rating = int(input("Enter rating: "))

bonus_calculation(salary, years, rating)
```

scholarship.py

```
def scholarship_details(score, family_income, attendance_percent):
    if score >= 85 and family_income <= 200000 and attendance_percent >= 90:
        print("Full Scholarship")

    elif score >= 75 and family_income <= 300000 and attendance_percent >=
80:
        print("Partial Scholarship")

    elif score >= 60 and family_income <= 400000 and attendance_percent >=
75:
        print("Financial Assistance Available")

    else:
        print("Not Eligible")


student_marks = int(input("Enter marks: "))
annual_income = int(input("Enter family income: "))
attendance_value = int(input("Enter attendance: "))

scholarship_details(student_marks, annual_income, attendance_value)
```

---

6. Patterns & Star Programs

pattern.py

```
print("Pattern 1")
for row in range(1, 6):
    print("*" * row)
```

```python
print("\nPattern 2")
for row in range(5, 0, -1):
    print("*" * row)

print("\nPattern 3")
for row in range(1, 6):
    for column in range(1, row + 1):
        print(column, end="")
    print()

print("\nPattern 4")
for row in range(1, 6):
    print(str(row) * row)

print("\nPattern 5")
size = 4
for row in range(size):
    print(" " * (size - row - 1) + "*" * (2 * row + 1))

print("\nPattern 6")
for row in range(5):
    print("*****")
```

---

7. Recursion & Algorithms

fib.py

```python
def fibonacci(num):
    if num == 0:
        return 0
    elif num == 1:
        return 1
    else:
        return fibonacci(num - 1) + fibonacci(num - 2)

limit = int(input("Enter terms: "))

for value in range(limit):
    print(fibonacci(value), end=" ")

print("\n================================")


def factorial(number):
    result = 1
    for item in range(1, number + 1):
        result *= item
    return result
```

```
num = int(input("Enter value: "))
print("Factorial =", factorial(num))

print("================================")

number = int(input())
reverse = 0

while number > 0:
    reverse = reverse * 10 + number % 10
    number //= 10

print(reverse)
```

prime.py

```
def check_prime(value):
    if value < 2:
        return False

    for item in range(2, value):
        if value % item == 0:
            return False

    return True


start_num = int(input("Enter starting number: "))
end_num = int(input("Enter ending number: "))

print("Prime Numbers:")

for number in range(start_num, end_num + 1):
    if check_prime(number):
        print(number, end=" ")

print("\n================================")


def multiplication_table(number):
    for value in range(1, 11):
        print(number, "x", value, "=", number * value)

num = int(input("Enter number: "))
multiplication_table(num)
```

8. Lists

operations.py

```
values = [5, 10, 15, 20]
print("Original List:", values)

print("First Element:", values[0])

values[1] = 12
print("Updated List:", values)

values.append(25)
print("After Append:", values)

values.insert(2, 8)
print("After Insert:", values)

values.remove(20)
print("After Remove:", values)

values.pop(0)
print("After Pop:", values)

print("Length:", len(values))

values.sort()
print("Sorted:", values)

values.reverse()
print("Reversed:", values)

print("Is 12 present?", 12 in values)

print("Traversing List")
for item in values:
    print(item, end=" ")
```

list1.py

```
items = [11, 22, 33, 44, 55]
print("Original:", items)

items.append(66)
print("After Append:", items)

items.insert(2, 27)
print("After Insert:", items)
```

```
items.remove(44)
print("After Remove:", items)

print("Length:", len(items))

items.sort()
print("Sorted List:", items)
```

---

9. Dictionaries & Tuples

dictio1.py

```
student_info = {
    "name": "Aman",
    "age": 19,
    "course": "CS"
}

print("Original Dictionary:", student_info)

student_info["city"] = "Mumbai"
print("After Adding City:", student_info)

student_info["age"] = 20
print("After Updating Age:", student_info)

student_info.pop("course")
print("After Removing Course:", student_info)

print("Student Name:", student_info["name"])
```

tuple.py

```
numbers = (5, 10, 15, 20, 25)

print("Tuple:", numbers)
print("First:", numbers[0])
print("Last:", numbers[-1])
print("Slice:", numbers[0:3])
print("Length:", len(numbers))

extra = (30, 35)
print("Concatenation:", numbers + extra)
print("Repeat:", numbers * 2)

print("Is 15 present?", 15 in numbers)
```

```
for value in numbers:
    print(value)

print("Maximum:", max(numbers))
print("Minimum:", min(numbers))
```

---

10. File Handling

filehandling.py

```
student_name = input("Enter student name: ")
student_marks = input("Enter marks: ")

file_obj = open("record.txt", "w")
file_obj.write("Student Information\n")
file_obj.write("-------------------\n")
file_obj.write("Name: " + student_name + "\n")
file_obj.write("Marks: " + student_marks + "\n")
file_obj.close()

print("Data Written Successfully")

grade = input("Enter grade: ")

file_obj = open("record.txt", "a")
file_obj.write("Grade: " + grade + "\n")
file_obj.close()

print("Data Appended Successfully")

file_obj = open("record.txt", "r")
content = file_obj.read()

print("File Content")
print(content)

file_obj.close()
```

---

11. NumPy

arr.py

```
import numpy as np

first_array = np.array([5, 10, 15])
```

```
second_array = np.array([20, 25, 30])

print("First Array:", first_array)
print("Second Array:", second_array)
```

one.py

```
import numpy as np

sample = np.array([2, 4, 6, 8])
print("Array:", sample)

matrix = np.zeros((3, 2))
print(matrix)
```

---

12. Pandas & Data Analysis

pandas1.py

```
import pandas as pd

student_data = {
    "Name": ["Riya", "Kunal", "Meena", "Ajay", "Neha"],
    "Age": [21, 22, 20, 23, 21],
    "Marks": [88, 75, 91, 84, 79],
    "City": ["Pune", "Delhi", "Mumbai", "Pune", "Nagpur"]
}

frame = pd.DataFrame(student_data)

print("DataFrame:\n", frame)

print("\nFirst Rows:\n", frame.head(3))
print("\nLast Rows:\n", frame.tail(2))

print("\nStatistics:\n", frame.describe())

print("\nNames:\n", frame["Name"])

print("\nSelected Columns:\n", frame[["Name", "Marks"]])

print("\nRows with Marks > 80:\n", frame[frame["Marks"] > 80])

frame["Result"] = ["Pass", "Pass", "Pass", "Pass", "Pass"]
print("\nUpdated DataFrame:\n", frame)

frame.drop("Result", axis=1, inplace=True)
```

```
print("\nSorted by Marks:\n", frame.sort_values(by="Marks"))
```

---

13. Matplotlib — Graphs & Charts

graph.py

```
import matplotlib.pyplot as plt

months = ['Jan', 'Feb', 'Mar', 'Apr', 'May']
profits = [120, 180, 220, 200, 260]

plt.figure(1)
plt.plot(months, profits)
plt.title("Monthly Profit - Line Graph")
plt.xlabel("Months")
plt.ylabel("Profit")

plt.figure(2)
plt.bar(months, profits)
plt.title("Monthly Profit - Bar Graph")
plt.xlabel("Months")
plt.ylabel("Profit")

plt.figure(3)
plt.pie(profits, labels=months, autopct='%1.1f%%')
plt.title("Profit Distribution")

plt.show()
```

scatter.py

```
import matplotlib.pyplot as plt

months = ['Jan', 'Feb', 'Mar', 'Apr', 'May']
revenue = [100, 140, 190, 210, 260]

students = [1, 2, 3, 4, 5]
marks = [55, 72, 80, 68, 95]

subjects = ['Math', 'Science', 'English']
subject_marks = [88, 92, 78]

plt.plot(months, revenue)
plt.title("Line Graph")
plt.show()
```

```
plt.bar(months, revenue)
plt.title("Bar Graph")
plt.show()

plt.pie(revenue, labels=months, autopct='%1.1f%%')
plt.title("Pie Chart")
plt.show()

plt.scatter(students, marks)
plt.title("Scatter Plot")
plt.show()

plt.hist(marks)
plt.title("Histogram")
plt.show()

plt.barh(subjects, subject_marks)
plt.title("Horizontal Bar Graph")
plt.show()
```